Experiment ‘p_5_9_f_cone_surface’ Results

June 5, 2026

Experiment outcome: FAILURE_VERIFY
Bad responses: 1
Responses containing `assume` : 0
Responses containing `expect` : 0
Resolution attempts: 3
Hard fails (resolution): 0
Soft fails (resolution): 0
Verification attempts: 3

Problem Specification

Problem name: p_5_9_f_cone_surface
Natural language statement: null
Method signature: p_5_9_f_cone_surface(r: real, h: real) returns (surface: real)

Ensures

- `-0.01 < (((surface / (3.14159 * r)) - r) * ((surface / (3.14159 * r)) - r)) - ((h * h) + (r * r)) < 0.01`

Requires

- `r > 0.0`

GenAI interactions

Below you will find all interactions between the 'user' (program) and the 'assistant' (OpenAI).

Program → GenAI

```
You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

p_5_9_f_cone_surface(r: real, h: real) returns (surface: real)

The method should respect the following contract:

requires r > 0.0, ensures -0.01 < (((surface / (3.14159 * r)) - r) * ((surface / (3.14159
    ↪  * r)) - r)) - ((h * h) + (r * r)) < 0.01

Produce and show only the Dafny body of this method, including the curly braces that
    ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
    ↪  'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-DnRBpi7gb2OvEuX5vNALyr4mg5DOd
Model: gpt-5.5-2026-04-23
Created at: 1780674961
Finish reason: stop
Usage: 362 tokens in, and 4644 tokens out

```
{
  assert h * h >= 0.0;
  assert r * r > 0.0;
  assert h * h + r * r >= 0.0;

  var x: real :| x * x == h * h + r * r;
  var c := 3.14159 * r;
  assert c > 0.0;
  assert c != 0.0;

  surface := c * (r + x);

  assert c == 3.14159 * r;
  assert surface / c == r + x;
  assert surface / (3.14159 * r) == r + x;
  assert (surface / (3.14159 * r)) - r == x;
  assert (((surface / (3.14159 * r)) - r) * ((surface / (3.14159 * r)) - r)) - ((h * h) +
      ↪  (r * r)) == 0.0;
  assert -0.01 < 0.0;
  assert 0.0 < 0.01;
```

```
}
```

Program → GenAI

```
When using dafny verify, the below error is emitted and verify fails:

problem_data/p_5_9_f_cone_surface/program.dfy(9,14): Error: cannot establish the
    ↪ existence of LHS values that satisfy the such-that predicate. Note, no trigger was
    ↪  found for the such-that predicate, which may be the reason the proof failed. To
    ↪ give a trigger explicitly, use the {:trigger} attribute. For more information, see
    ↪  the section on quantifier instantiation rules in the reference manual.
  |
9 | var x: real :| x * x == h * h + r * r;
  | ^^^^^^^^^^^^^^^^^^^^^^^^^^


Dafny program verifier finished with 1 verified, 1 error

Correct the error in code by altering only the method body. Produce and show only the
    ↪ Dafny body, including the curly braces that surround it. Do not show the signature
    ↪  nor contract. You must not use 'assume' nor 'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-DnRE0RHbZADyvRXp9t6lgTRnmDoUY
Model: gpt-5.5-2026-04-23
Created at: 1780675096
Finish reason: stop
Usage: 872 tokens in, and 7799 tokens out

```
{
  var S := h * h + r * r;
  assert h * h >= 0.0;
  assert r * r > 0.0;
  assert S >= 0.0;

  var lo := 0.0;
  var hi := S + 1.0;

  assert lo * lo <= S;
  assert S * S >= 0.0;
  calc {
    hi * hi - S;
    ==
    (S + 1.0) * (S + 1.0) - S;
    ==
    S * S + S + 1.0;
  }
  assert hi * hi - S >= 0.0;
  assert S <= hi * hi;
```

```
while 0.01 <= (hi - lo) * (hi + lo)
  invariant 0.0 <= lo <= hi
  invariant lo * lo <= S <= hi * hi
  decreases hi - lo
{
  var oldLo := lo;
  var oldHi := hi;
  var width := oldHi - oldLo;

  assert width >= 0.0;
  assert oldHi + oldLo >= 0.0;
  if width == 0.0 {
    assert (hi - lo) * (hi + lo) == 0.0;
    assert false;
  }
  assert width > 0.0;

  var mid := (oldLo + oldHi) / 2.0;
  assert mid - oldLo == width / 2.0;
  assert oldHi - mid == width / 2.0;
  assert width / 2.0 > 0.0;
  assert oldLo < mid < oldHi;

  if mid * mid <= S {
    lo := mid;
    assert hi == oldHi;
    assert lo * lo <= S;
    assert S <= hi * hi;
    assert hi - lo == width / 2.0;
    assert hi - lo < width;
    assert hi - lo < oldHi - oldLo;
  } else {
    hi := mid;
    assert lo == oldLo;
    assert S <= hi * hi;
    assert lo * lo <= S;
    assert hi - lo == width / 2.0;
    assert hi - lo < width;
    assert hi - lo < oldHi - oldLo;
  }
}

assert (hi - lo) * (hi + lo) < 0.01;
calc {
  hi * hi - lo * lo;
  ==
  (hi - lo) * (hi + lo);
}
assert S - lo * lo <= hi * hi - lo * lo;
assert S - lo * lo < 0.01;
```

```
  assert lo * lo - S <= 0.0;
  assert -0.01 < lo * lo - S;
  assert lo * lo - S < 0.01;

  var den := 3.14159 * r;
  assert den > 0.0;
  assert den != 0.0;

  surface := den * (r + lo);

  assert den == 3.14159 * r;
  assert surface / den == r + lo;
  assert surface / (3.14159 * r) == r + lo;
  assert (surface / (3.14159 * r)) - r == lo;
  assert (((surface / (3.14159 * r)) - r) * ((surface / (3.14159 * r)) - r)) - ((h * h) +
      ↪  (r * r)) == lo * lo - S;
  assert -0.01 < (((surface / (3.14159 * r)) - r) * ((surface / (3.14159 * r)) - r)) - ((
      ↪ h * h) + (r * r)) < 0.01;
}
```

Program → GenAI

```
When using dafny verify, the below error is emitted and verify fails:

problem_data/p_5_9_f_cone_surface/program.dfy(25,2): Error: decreases expression might
    ↪ not decrease
   |
25 | while 0.01 <= (hi - lo) * (hi + lo)
   | ^^^^^



Dafny program verifier finished with 1 verified, 1 error

Correct the error in code by altering only the method body. Produce and show only the
    ↪ Dafny body, including the curly braces that surround it. Do not show the signature
    ↪  nor contract. You must not use 'assume' nor 'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-DnRHisdfMwnmGFRCaFyIhsH8y7tmA
Model: gpt-5.5-2026-04-23
Created at: 1780675326
Finish reason: stop
Usage: 1866 tokens in, and 7970 tokens out

```
{
  var S := h * h + r * r;
  assert h * h >= 0.0;
  assert r * r > 0.0;
  assert S >= 0.0;
```

```
var lo := 0.0;
var hi := S + 1.0;

assert lo * lo <= S;
assert S * S >= 0.0;
calc {
  hi * hi - S;
  ==
  (S + 1.0) * (S + 1.0) - S;
  ==
  S * S + S + 1.0;
}
assert hi * hi - S >= 0.0;
assert S <= hi * hi;

while 0.01 <= (hi - lo) * (hi + lo)
  invariant 0.0 <= lo <= hi
  invariant lo * lo <= S <= hi * hi
  decreases hi - lo
{
  var oldLo := lo;
  var oldHi := hi;
  var width := oldHi - oldLo;

  assert width >= 0.0;
  assert oldHi + oldLo >= 0.0;
  if width == 0.0 {
    assert (hi - lo) * (hi + lo) == 0.0;
    assert false;
  }
  assert width > 0.0;

  var mid := (oldLo + oldHi) / 2.0;
  assert mid - oldLo == width / 2.0;
  assert oldHi - mid == width / 2.0;
  assert width / 2.0 > 0.0;
  assert oldLo < mid < oldHi;

  if mid * mid <= S {
    lo := mid;
    assert hi == oldHi;
    assert lo * lo <= S;
    assert S <= hi * hi;
    assert hi - lo == width / 2.0;
  } else {
    hi := mid;
    assert lo == oldLo;
    assert S <= hi * hi;
    assert lo * lo <= S;
    assert hi - lo == width / 2.0;
```

```
    }

    assert hi - lo == width / 2.0;
    assert width / 2.0 < width;
    assert hi - lo < width;
    assert width == oldHi - oldLo;
    assert hi - lo < oldHi - oldLo;
  }

  assert (hi - lo) * (hi + lo) < 0.01;
  calc {
    hi * hi - lo * lo;
    ==
    (hi - lo) * (hi + lo);
  }
  assert S - lo * lo <= hi * hi - lo * lo;
  assert S - lo * lo < 0.01;
  assert lo * lo - S <= 0.0;
  assert -0.01 < lo * lo - S;
  assert lo * lo - S < 0.01;

  var den := 3.14159 * r;
  assert den > 0.0;
  assert den != 0.0;

  surface := den * (r + lo);

  assert den == 3.14159 * r;
  assert surface / den == r + lo;
  assert surface / (3.14159 * r) == r + lo;
  assert (surface / (3.14159 * r)) - r == lo;
  assert (((surface / (3.14159 * r)) - r) * ((surface / (3.14159 * r)) - r)) - ((h * h) +
      ↪  (r * r)) == lo * lo - S;
  assert -0.01 < (((surface / (3.14159 * r)) - r) * ((surface / (3.14159 * r)) - r)) - ((
      ↪ h * h) + (r * r)) < 0.01;
}
```

Program → GenAI

```
When using dafny verify, the below error is emitted and verify fails:

problem_data/p_5_9_f_cone_surface/program.dfy(25,2): Error: decreases expression might
    ↪ not decrease
   |
25 | while 0.01 <= (hi - lo) * (hi + lo)
   | ^^^^^


Dafny program verifier finished with 1 verified, 1 error
```

```
Correct the error in code by altering only the method body. Produce and show only the
    ↪ Dafny body, including the curly braces that surround it. Do not show the signature
    ↪  nor contract. You must not use 'assume' nor 'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-DnRLqsgkxar9sxqft6xNeMmYHIH7S
Model: gpt-5.5-2026-04-23
Created at: 1780675582
Finish reason: length
Usage: 2941 tokens in, and 6487 tokens out
The below is a raw response from GenAI that could not be parsed. It is likely that GenAI got 'stuck' in a loop and hit the pre-set token limit (i.e., finish reason is 'length').

Final Program

```
method p_5_9_f_cone_surface(r: real, h: real) returns (surface: real)
       requires r > 0.0
       ensures -0.01 < (((surface / (3.14159 * r)) - r) * ((surface / (3.14159 * r)) - r)
           ↪ ) - ((h * h) + (r * r)) < 0.01
{
  var S := h * h + r * r;
  assert h * h >= 0.0;
  assert r * r > 0.0;
  assert S >= 0.0;

  var lo := 0.0;
  var hi := S + 1.0;

  assert lo * lo <= S;
  assert S * S >= 0.0;
  calc {
    hi * hi - S;
    ==
    (S + 1.0) * (S + 1.0) - S;
    ==
    S * S + S + 1.0;
  }
  assert hi * hi - S >= 0.0;
  assert S <= hi * hi;

  while 0.01 <= (hi - lo) * (hi + lo)
    invariant 0.0 <= lo <= hi
    invariant lo * lo <= S <= hi * hi
    decreases hi - lo
  {
    var oldLo := lo;
    var oldHi := hi;
```

```
  var width := oldHi - oldLo;

  assert width >= 0.0;
  assert oldHi + oldLo >= 0.0;
  if width == 0.0 {
    assert (hi - lo) * (hi + lo) == 0.0;
    assert false;
  }
  assert width > 0.0;

  var mid := (oldLo + oldHi) / 2.0;
  assert mid - oldLo == width / 2.0;
  assert oldHi - mid == width / 2.0;
  assert width / 2.0 > 0.0;
  assert oldLo < mid < oldHi;

  if mid * mid <= S {
    lo := mid;
    assert hi == oldHi;
    assert lo * lo <= S;
    assert S <= hi * hi;
    assert hi - lo == width / 2.0;
  } else {
    hi := mid;
    assert lo == oldLo;
    assert S <= hi * hi;
    assert lo * lo <= S;
    assert hi - lo == width / 2.0;
  }

  assert hi - lo == width / 2.0;
  assert width / 2.0 < width;
  assert hi - lo < width;
  assert width == oldHi - oldLo;
  assert hi - lo < oldHi - oldLo;
}

assert (hi - lo) * (hi + lo) < 0.01;
calc {
  hi * hi - lo * lo;
  ==
  (hi - lo) * (hi + lo);
}
assert S - lo * lo <= hi * hi - lo * lo;
assert S - lo * lo < 0.01;
assert lo * lo - S <= 0.0;
assert -0.01 < lo * lo - S;
assert lo * lo - S < 0.01;

var den := 3.14159 * r;
assert den > 0.0;
```

```
  assert den != 0.0;

  surface := den * (r + lo);

  assert den == 3.14159 * r;
  assert surface / den == r + lo;
  assert surface / (3.14159 * r) == r + lo;
  assert (surface / (3.14159 * r)) - r == lo;
  assert (((surface / (3.14159 * r)) - r) * ((surface / (3.14159 * r)) - r)) - ((h * h) +
      ↪  (r * r)) == lo * lo - S;
  assert -0.01 < (((surface / (3.14159 * r)) - r) * ((surface / (3.14159 * r)) - r)) - ((
      ↪ h * h) + (r * r)) < 0.01;
}
```

Total Token Usage

Input tokens: 6041
Output tokens: 26900
Reasoning tokens: 24576
Sum of 'total tokens': 32941

Experiment Timings

Overall Experiment started at 1780674961074, ended at 1780675742234, lasting 781160ms (781.16 seconds)
Iteration #4 started at 1780675581546, ended at 1780675742234, lasting 160688ms (160.69 seconds)
Iteration #1 started at 1780674961074, ended at 1780675096472, lasting 135398ms (135.40 seconds)
Iteration #2 started at 1780675096477, ended at 1780675322291, lasting 225814ms (225.81 seconds)
Iteration #3 started at 1780675322291, ended at 1780675581546, lasting 259255ms (259.26 seconds)